

Gene expression pattern

Nrk: a murine X-linked *NIK* (Nck-interacting kinase)-related kinase gene expressed in skeletal muscle

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Abstract

We report the cloning and expression pattern of a novel Ste20-type kinase gene, *NIK*-related kinase (*Nrk*), located on the mouse X chromosome. The full-length *Nrk* cDNA encodes a 1455-amino-acid polypeptide characterized by a N-terminal Ste20-type catalytic domain and a C-terminal regulatory domain characteristic of the group I GCK subfamily. The overall structure of the *NRK* protein is closely related to that of Nck-interacting kinase (*Nik*). In situ hybridization revealed that *Nrk* was predominantly expressed in skeletal muscle during mouse embryogenesis. *Nrk* gene expression was detected in the myotome at 10.5 dpc and, thereafter, was observed in developing skeletal musculature from 11.5 to 13.5 dpc. However, expression in skeletal muscle was not observed in adults. © 1999 Elsevier Science Ireland Ltd. All rights reserved.

Keywords: *Nrk*; Ste20; *NIK*; GCK; KHS; HPK1; Kinase; Skeletal muscle; Myogenesis; Myogenin

1. Results and discussion

The Ste20-type kinase family, which is believed to regulate MAP kinase cascades (Cobb et al., 1994; Schlesinger et al., 1998; Liu et al., 1999), is divided into three subfamilies in higher eukaryotes based on its overall protein structure: (1) PAK subfamily characterized by a p21-binding domain; (2) Group I germinal center kinase (GCK) subfamily characterized by a C-terminal regulatory domain; and (3) Group II GCK subfamily which contains neither of the above domains (Kyriakis, 1999). Many members of the group I GCK subfamily have been shown to activate the JNK/SAPK pathway of MAP kinase cascades (Pombo et al., 1995; Hu et al., 1996; Kiefer et al., 1996; Su et al., 1997; Tung and Blenis, 1997).

In this study, we cloned a full-length cDNA encoding a novel Ste20-type kinase from a 11.5 dpc mouse cDNA library (accession number AB020741). Its deduced amino acid structure is characterized by: (1) a Ste20-type catalytic domain at N-terminal region; (2) an intermediate region which contains one coiled-coil structure and five putative

SH3-binding sites; and (3) a C-terminal domain conserved among members of the group I GCK subfamily such as Nck-interacting kinase (*NIK*) (Su et al., 1997), GCK (Katz et al., 1994), kinase homologous to SPS1/STE20 (KHS; Tung and Blenis, 1997), and hematopoietic progenitor kinase (HPK; Hu et al., 1996) (Fig. 1A,B). Due to the structural similarities with *NIK*, we named this gene *NIK*-related kinase (*Nrk*). Linkage analysis showed that *Nrk* is located on the mouse X chromosome and is 1.9 cM distal to DXMit3. These findings indicate that *Nrk* is a novel X-linked *NIK*-related kinase gene belonging to the group I GCK subfamily.

Many members of the Ste20 family, including *NIK* and *GCK*, have been reported to be widely expressed in various tissues and cell types (Katz et al., 1994; Su et al., 1997; Schinkmann and Blenis, 1997; Pytowski et al., 1998). In contrast, whole mount in situ hybridization analysis revealed that *Nrk* was predominantly expressed in skeletal muscle during mouse embryogenesis. *Nrk* expression was not observed between 7.5 and 9.5 dpc, although *Myogenin*, a myogenic transcription factor, was activated around at 8.5–9.5 dpc (Fig. 2A,B). *Nrk* expression was first observed in the ventral myotome at 10.5 dpc (Fig. 2C–J). At 11.5 dpc, strong expression of *Nrk* was observed in the myotome

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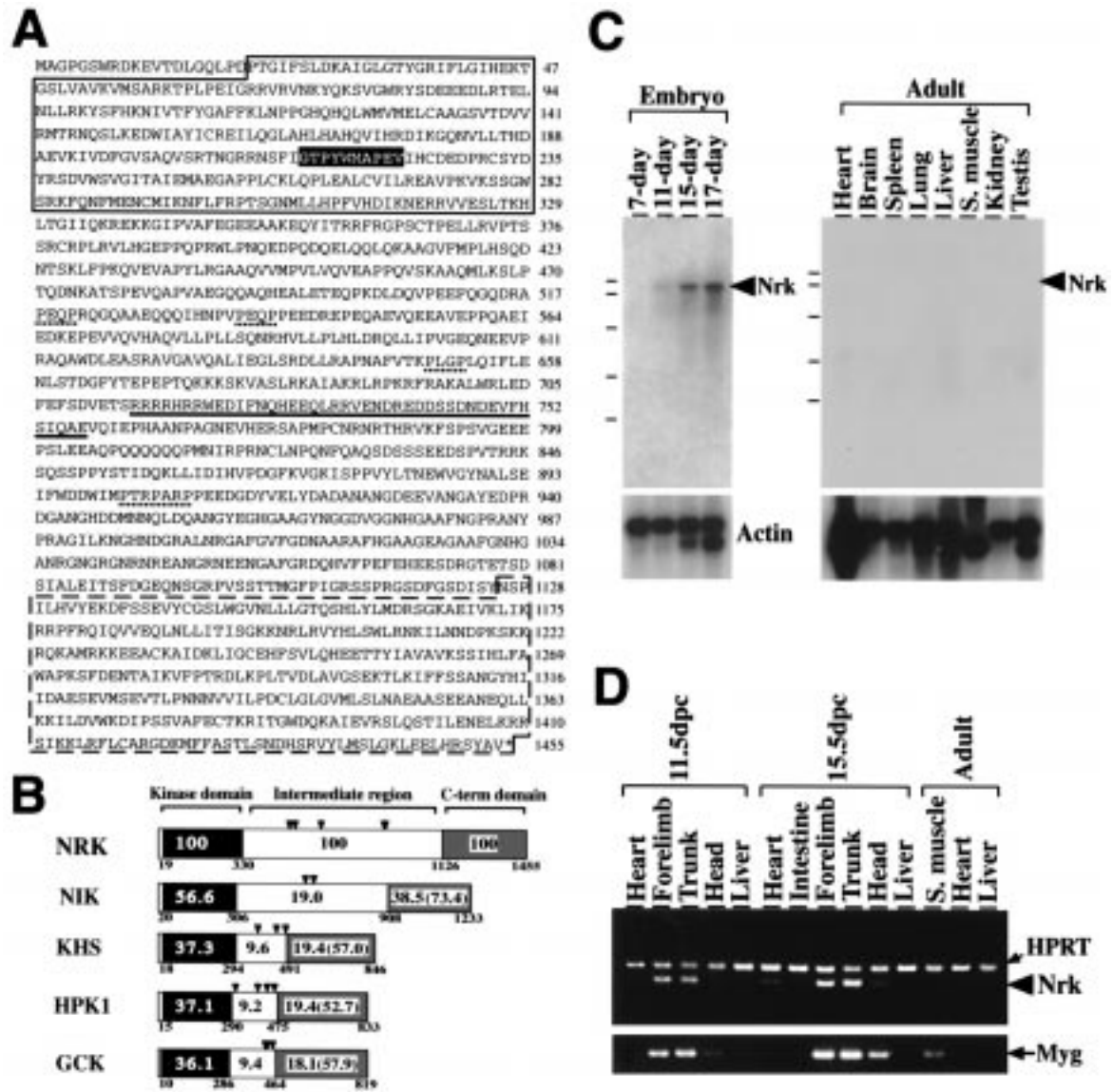
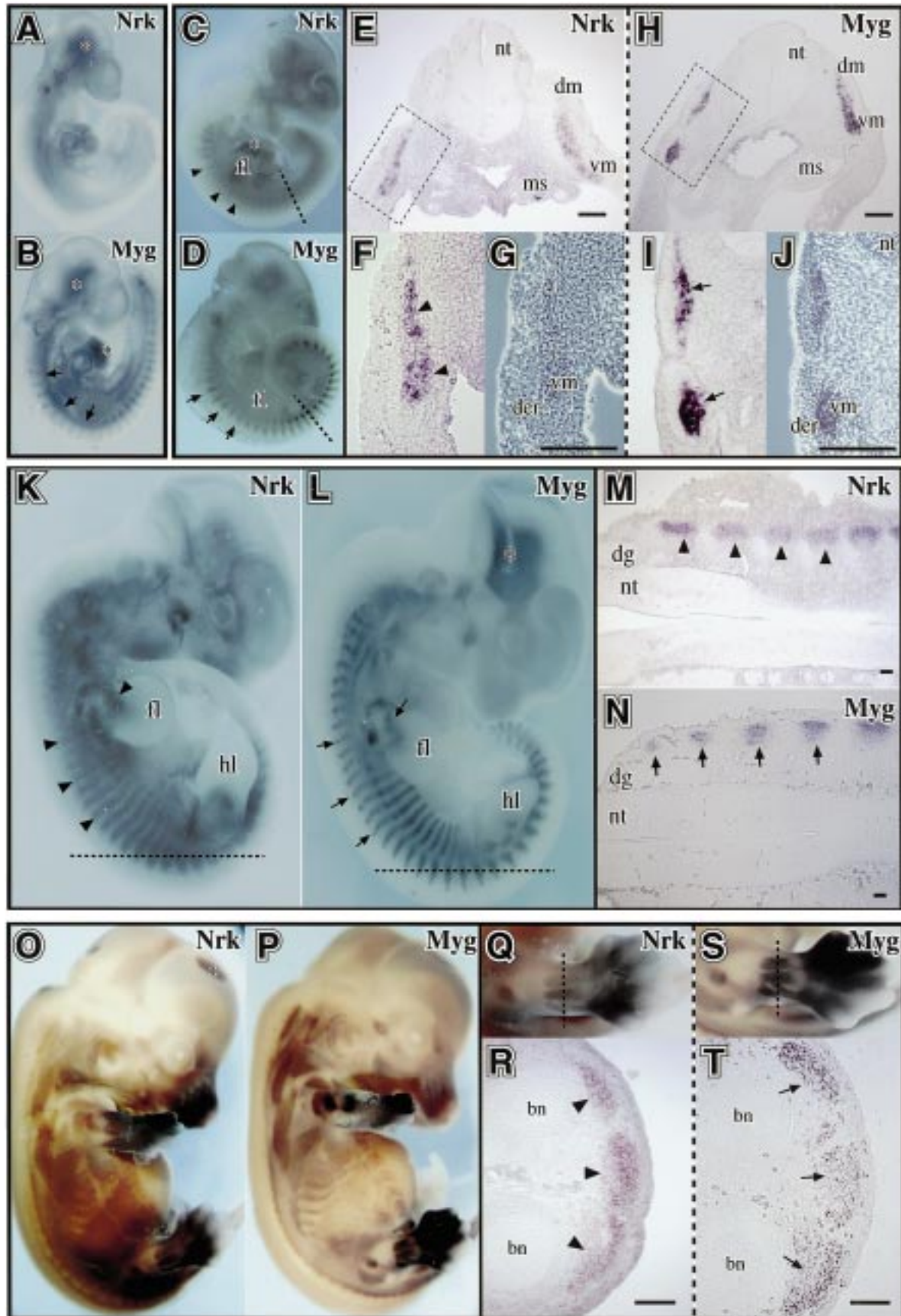


Fig. 1. (A) Deduced amino acid sequences of mouse *Nrk*. The catalytic and regulatory domains are surrounded by solid and broken lines, respectively. The catalytic domain of *Nrk* contains the sequence GTPY/FWMAPEV characteristic of the Ste20 family and is indicated by a solid box. The putative SH3-binding sites of a PXXP motif are indicated by a broken underline. The coiled-coil structure examined by the method of Lupas et al. (1991) is marked by a bold underline. (B) Schematic comparison of the protein features of NRK, NIK, KHS, HPK1, and GCK. The positions of the putative SH3-binding sites are shown by arrowheads. The number within each box indicates the ratio (%) of identical amino acids relative to *Nrk* (the number inside parenthesis shows the similarity relative to NRK). (C) Northern blot analysis showing *Nrk* expression (approximately 8.0 kb transcript) in embryos at various stages and in various tissues of adult mice. RNA markers are shown as bars at the left of panel A (9.5, 7.5 4.4, 2.4, and 1.35 kb from top to bottom). The lower panel shows the same blot hybridized with a β -actin probe as a control for RNA loading. (D) RT-PCR analysis showing the expression of *Nrk* (arrowheads in the upper panel) and *Myogenin* (lower panel). HPRT was used as an internal control (arrow in the upper panel). All RT-PCR products of *Nrk* were amplified with primers that span exon boundaries, thereby controlling for genomic DNA contamination of RNA samples. No amplification products were detected in samples from reactions without reverse transcriptase (not shown).

along the entire anterior-posterior axis, as well as in developing limb buds (Fig. 2K–N). *Nrk* expression domains in the trunk and limbs were similar to, but extended over a wider range than, those of *Myogenin*-expressing cells. Such find-

Fig. 2. Whole mount in situ hybridization analysis of *Nrk* expression in mouse embryos. Expression of *Nrk* and *Myogenin* (*Myg*) at 9.5 (A,B), 10.5 (C–J), 11.5 (K–N), and 13.5 (O–T) dpc was examined separately using gene-specific DIG-labeled RNA probes. In plates (C,D,K,L,Q, and S), the broken lines indicate the sectioning positions of the embryos shown in plates (E–G,H–J,M,N,R, and T), respectively. Plates F and I show a higher magnification of the area indicated by the broken rectangle in plates E and H, respectively. Plates G and J display a phase contrast image of plates F and I, respectively. Arrowheads or arrows show the localization of *Nrk* or *Myogenin* transcripts. Bars at the right lower edge indicate 200 μ m. fl, forelimb buds; hl, hindlimb buds; nt, neural tube; dm, dorsal myotome; vm, ventral myotome; ms, mesonephros; dg, dorsal root ganglion; bn, bone; asterisk, non-specific signals due to antibody trapping in lumen.



ings suggest that *Nrk* expression may be not restricted to the terminal differentiation step of myogenesis, although comparative analysis using a double-labeling technique with various muscle markers should be required for further study. From 12.5 to 13.5 dpc, *Nrk* expression was observed throughout the developing skeletal musculature within the trunk, limbs, neck and cheek (Fig. 2O–T).

Northern blot analysis showed that *Nrk* was transcribed in the 11 dpc embryo and that expression increased from 15 to 17 dpc. This expression may coincide with the relative increase of skeletal muscle during embryogenesis (Fig. 1C). Semi-quantitative RT-PCR analysis also confirmed that *Nrk* is expressed in tissues containing skeletal muscle (e.g. limb buds, trunk) of 11.5 and 15.5 dpc embryos; however, only low level expression was detected in the heart of the 15.5 dpc embryo (Fig. 1D). In contrast, *Nrk* expression was not detected in any of the adult tissues examined, including skeletal muscle (Fig. 1C,D). These results demonstrate that the muscle-predominant expression of *Nrk* during embryogenesis is down-regulated as development proceeds.

Recently, *misshapen* in *Drosophila* and *mig15* in *C. elegans*, both structurally similar to mammalian *NIK*, have been shown to be essential for dorsal closure, and Q-neuroblast migration and muscle arm targeting, respectively (Su et al., 1998). Since *Nrk* is also a novel *NIK*-related kinase gene with a skeletal muscle-predominant expression, further characterization of this gene may shed light on signal transduction in the development and maturation of skeletal muscle.

2. Methods

2.1. Isolation of *Nrk* cDNA and genetic mapping

We isolated *Nrk* cDNA fragments by a PCR-based approach using degenerate primers for the kinase domain (Wilks, 1989) and further isolated the full length cDNA clone from 11.5 dpc mouse cDNA Lambda gt11 and gt10 libraries (CLONTECH). Linkage mapping was carried out using an intersubspecific backcross panel constructed with C57BL/6J and *Mus m. molossinus*, as previously described (Okamoto et al., 1995; Suhara et al., 1996).

2.2. RNA expression analysis

cDNA fragments (1020 bp) corresponding to the *Nrk* intermediate region were used as a *Nrk*-specific probe. The *Myogenin*-specific probe (Fujisawa-Sehara et al., 1990) was kindly provided by Dr A. Fujisawa-Sehara. Whole-mount in situ hybridization was performed as described by Wilkinson and Nieto (1993). After whole-mount in situ hybridization, embryos were sectioned at 10–20 μ m thickness. Multiple tissue Northern blots, each containing 2 μ g-poly(A)⁺ RNA per sample (CLONTECH), were hybridized with ³²P-labeled probes as described

previously (Kanai et al., 1996). cDNA samples were transcribed from 1 μ g of total RNA and amplified via PCR using *Nrk*-specific primers (Forward: 5'-CTTGACAAA-CATCTTACTGGGATC-3'; Reverse: 5'-TGTATTGT-CCTGAGAATGTA GAGGC-3'), *HPRT* primers (Kanai-Azuma et al., 1997), and *Myogenin* primers (Yagami-Hiro-masa et al., 1995).

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